When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models

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Abstract

Purpose: Graph neural networks (GNNs) are increasingly proposed for healthcare insurance fraud because collusive provider rings leave traces in the patient–provider network, but it is unclear when they improve on tabular models. **Methods:** We built a synthetic claims benchmark in which the feature-level signal of ring claims and the density of the ring's doctor–patient block are varied independently, every test ring is formed by doctors and patients never seen in training, and each audit period is analysed separately to avoid leakage. We compared 14 methods—tabular models with and without engineered graph features, an MLP control, five heterogeneous GNNs (GraphSAGE, RGCN, HAN, Simple-HGN, HGT), two hybrids and an unsupervised overlap score—with equal tuning budgets over nine regimes and 54 independently generated worlds, and validated on a public provider-labelled Medicare-style dataset (5,410 providers). **Results:** Without a feature-level signal, claim-only models were at chance, while per-entity aggregates detected most rings (average precision 0.89–0.93). Structure added up to 0.10 average precision, and only when ring members were moderately or densely interlinked. Pooled over all worlds, no GNN outperformed gradient boosting with engineered graph features (GraphSAGE −0.0001, HGT −0.011); hybrids gained +0.008. GNNs were 0.08–0.15 worse on isolated fraud,

HGT lost most of its ring detection without degree encodings, and letting test rings share members with training rings inflated performance by up to 0.60. On real data, gradient boosting on provider aggregates was best (0.704); graph features and GNNs did not help. **Conclusion:** Relational modelling pays off only for dense collusion and, even then, mainly through structure that simple engineered features capture. Graph models should be evaluated against graph-augmented tabular baselines on unseen entities.

Keywords: Healthcare fraud detection, provider rings, heterogeneous graph neural networks, gradient boosting, synthetic benchmark, data leakage, evaluation methodology

1 Introduction

Health insurance fraud causes losses estimated in the tens of billions of dollars each year in the United States alone [1]. It takes two broad forms. In *isolated* fraud a single provider inflates or fabricates individual claims, so the evidence lies in the claim itself: amounts, procedure counts or coding that are unusual for the provider's specialty. In *collusive* fraud a group of providers and recruited patients file claims together. Each claim can look ordinary; the evidence lies in who bills whom—patients who see many members of the same small group of providers, and providers who share an unusually large fraction of their patients.

The second form motivates graph neural networks (GNNs) over the patient–provider–facility network, in particular heterogeneous architectures such as RGCN [16], HAN [17], HGT [18] and Simple-HGN [19]. Whether GNNs are needed, however, is an empirical question. Gradient-boosted trees remain strong on tabular data [24], relational signals can be engineered as features, and benchmark studies outside healthcare report that tree ensembles with neighbourhood features often match or beat GNNs on graph anomaly detection [22, 23].

Answering the question for healthcare claims is difficult for two reasons. First, public claims data with labelled collusion are not available, and real labels conflate fraud typologies. Second, evaluation is fragile: identifiers that encode the label, statistics computed on the full dataset, and providers that appear in both training and test data inflate apparent performance [27, 28]. Fraud rings make the last problem acute, because a model can score a test claim correctly by recognising a provider it saw labelled during training instead of recognising coordination.

We therefore study the question in a controlled setting and check it on real data. Our contributions are:

1. **A regime-controllable benchmark.** A synthetic claims generator in which the feature-level signal of ring claims (δ) and the density of the ring's doctor–patient block (s) are varied independently, with an *emergent* protocol in which every test ring is formed by doctors and patients never seen in a training ring, and an audit-period design that keeps features and graphs free of future information (Section 3). Code, generator and tests are released.

2. **A controlled comparison** of 14 methods—tabular models with and without engineered graph features, an MLP control, five heterogeneous GNNs, two hybrids and an unsupervised overlap score—with an equal tuning budget, nine regimes, six independently generated worlds per regime and paired statistics (Sections 4 and 6).
3. **Mechanism and memorisation analyses**: degree-preserving rewiring, relation and encoding ablations, depth, and a comparison of new versus recurring rings that quantifies how much apparent ring detection comes from recognising known members (Section 7).
4. **Real-world validation** on a public, provider-labelled Medicare-style claims dataset with a provider-disjoint protocol (Section 5).

Our main findings are as follows. Claim features alone cannot reveal rings whose claims look legitimate, but per-entity aggregates capture most of the evidence, because ring members are unusually active. Coordination structure adds measurable value only when the ring's doctor–patient block is dense. Where it does, gradient boosting with a few engineered shared-patient features is as good as or better than every GNN we tested; GNNs that also receive those features gain a small but consistent amount (+0.008 AP pooled). GNNs are clearly worse on isolated fraud, because message passing dilutes claim-level anomalies, and attention-based GNNs lose most of their ring-detection ability without explicit degree encodings. Evaluations that let test rings share members with training rings overstate ring detection, by up to 0.60 AP for a model that only sees entity attributes. On the real data, where labels show no relational assortativity, no graph-based method improved on gradient boosting over provider aggregates.

2 Related Work

Tabular healthcare fraud detection.

Most published healthcare-fraud detectors are supervised classifiers over claim- or provider-level features. On Medicare data labelled with the federal exclusion list (LEIE), Bauder and Khoshgoftaar [2] compared classical learners, Herland et al. [3] combined several Medicare data sources, and Johnson and Khoshgoftaar [4] studied neural networks under extreme class imbalance. Earlier work on Medicaid data used multidimensional data models and outlier analysis [5]. These models score each claim or provider from its own attributes and pre-computed aggregates; relational information enters, if at all, through hand-engineered features.

Graph-based fraud and anomaly detection.

Surveys of graph-based anomaly detection [6, 7] document how coordination leaves structural traces such as dense blocks, lockstep behaviour and unusual egonets. Unsupervised detectors exploit these traces directly, e.g. OddBall [8], CopyCatch [9] and FRAUDAR [10]. Graph neural networks (GNNs) have been applied to review fraud [11], underground-forum key players [12], and camouflaged or class-imbalanced fraudsters [13, 14]; spectral analyses argue that anomalies shift graph spectra in ways that standard message passing can blur [15].

Heterogeneous GNNs.

Relational GCN [16] uses relation-specific weights; HAN [17] aggregates over hand-chosen meta-paths with node- and semantic-level attention; HGT [18] parameterises attention by node and edge type without meta-paths; and Simple-HGN [19] showed that a GAT with edge-type embeddings, residual attention and output normalisation is competitive with more elaborate heterogeneous architectures when baselines are tuned fairly.

When do graph models beat tabular models?

Evidence on this question is mixed. On the Elliptic Bitcoin dataset a random forest on node features outperformed graph convolutional networks under a temporal split [22]. GADBench [23] found that tree ensembles with simple neighbourhood aggregation are among the strongest supervised graph anomaly detectors across ten datasets, and tree ensembles remain hard to beat on tabular data generally [24]. Synthetic benchmarks with injected typologies make the question testable: Altman et al. [25] generate money-laundering patterns with known ground truth, and Jesus et al. [26] release controlled tabular fraud suites. To our knowledge no comparable controlled study exists for heterogeneous patient–provider–facility claim graphs, where the relevant typology is the collusive provider ring.

Evaluation pitfalls.

Leakage through features that encode the target, through preprocessing fitted on the full dataset, or through entities shared between training and test data inflates reported performance [27, 28]. GNN benchmarks are additionally sensitive to split choice and tuning effort [29]. These risks are acute for fraud rings, whose members recur across claims: a model can score a test claim correctly by recognising a provider it saw labelled in training rather than by recognising coordination.

3 Benchmark Design

3.1 Synthetic claims world

We simulate 6,000 patients, 300 doctors and 40 hospitals in five regions over four audit periods of six months (Fig. 1; Table A1). Each doctor has one of eight specialties, which sets the distribution of claim amounts, inpatient probability and procedure counts. Patients have heterogeneous utilisation (gamma-distributed claim rates with mean three claims per two years) and one to three regular doctors in their region; 85% of legitimate claims go to a regular doctor and the rest to a popularity-weighted random doctor. Claim amounts are log-normal given the specialty, procedure count and inpatient status.

Isolated fraud.

Three per cent of claims are upcoded claims filed through a patient's own regular doctor. Their log-amount, procedure count, inpatient probability and procedure-code

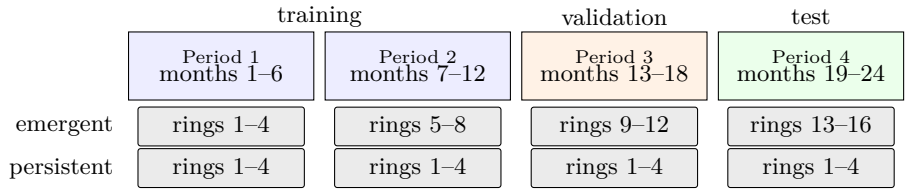


Fig. 1 Audit-period protocol. Each six-month period is analysed separately: features and graphs use only that period's claims. Under the emergent protocol every period has new rings formed by disjoint doctors and patients, so test rings are unseen; under the persistent protocol the same rings recur and can be recognised by their members.

level are shifted by two standard deviations; their structure is that of an ordinary claim.

Provider rings.

A ring consists of four colluding doctors and twelve recruited patients, drawn preferentially from high utilisers. During its active period the ring files 16 claims per month between linked doctor–patient pairs, clustered around monthly burst days. Two parameters are varied independently:

- the *feature shift* $\delta \in \{0, 1, 2\}$ applied to ring claims, in standard deviations of log-amount (with matching shifts in procedures, inpatient probability and coding level). At $\delta = 0$ a ring claim is drawn from exactly the distribution of a legitimate claim of the same doctor's specialty;
- the *structural density* $s \in \{0.1, 0.5, 0.9\}$, the probability that a ring patient is linked to each ring doctor. Ring claim volume is fixed, so s changes only how the claims are spread over the doctor–patient block (realised block densities ≈ 0.37, 0.62 and 0.79).

Ring patients keep a single regular doctor, so the number of distinct doctors they see is governed by s. Their elevated claim volume, a degree signal visible to every model, is the same in all regimes.

Protocols.

Under the *emergent* protocol (default), four new rings start in each period and rings in different periods are formed by disjoint doctors and patients, so every test ring is unseen during training. Under the *persistent* protocol the same four rings are active in all periods, which reproduces the setting of most prior synthetic evaluations and lets a model succeed by recognising known members. Both protocols have identical claim and ring-claim volumes per period. Ground truth (fraud type and ring identifier) is stored separately from the claim records and never enters feature construction.

3.2 Evaluation protocol

Periods 1–2 are used for training, period 3 for validation (early stopping, threshold selection) and period 4 for testing. Each period is analysed on its own, as in a retrospective audit: a claim's aggregate and graph features, and the graph a GNN operates

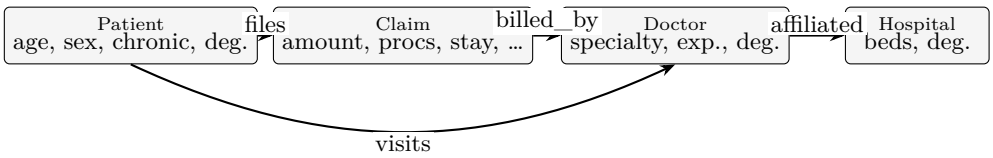


Fig. 2 Heterogeneous claim graph of one audit period. Every relation also has a reverse relation (not drawn), so every node type receives messages and doctor context reaches claim nodes. Entity nodes carry static attributes and log(1 + degree) per incident relation; claim nodes carry only claim-intrinsic attributes.

on, are built only from claims of the same period. This keeps every split's features on the same time scale and prevents information from later periods reaching earlier ones; it is verified by automated tests (Section 10).

4 Methods

4.1 Tabular features

- **C** (claim): log-amount, procedures, length of stay, inpatient flag, coding level, month (cyclical), patient age, sex and chronic-condition count, doctor experience and specialty, hospital size.
- **A** (aggregates within the period, no labels): the patient's claim count, distinct doctors and mean/maximum log-amount; the doctor's claim count, distinct patients and mean log-amount; hospital claim count; patient–doctor pair count; and the claim's log-amount relative to the patient, doctor and specialty means.
- **G** (graph features on the period's patient–doctor bipartite graph, no labels): the doctor's maximum shared-patient Jaccard similarity with another doctor and number of doctors sharing at least two patients, the mean degree of the doctor's patients, the patient's mean doctor overlap and share of doctor pairs that co-treat other patients, and the triadic closure of the claim (how many of the patient's other doctors share patients with the claim's doctor), raw and degree-normalised.

4.2 Heterogeneous graph

Each period's graph has patient, doctor, hospital and claim nodes and the relations *files* (patient→claim), *billed_by* (claim→doctor), *affiliated* (doctor→hospital) and *visits* (patient→doctor), each with a reverse relation so that every node type receives messages. Claim nodes carry claim-intrinsic attributes; entity nodes carry their static attributes and log(1 + degree) for each incident relation, a label-free structural encoding included because attention- and mean-based aggregators cannot otherwise count neighbours [21]. No engineered aggregates or labels are used as node features (Fig. 2). Standardisation statistics are fitted on training data.

4.3 Models

- **Tabular**: logistic regression (LR) and random forest (RF) on C+A; histogram gradient boosting (GB) [33] on C, C+A and C+A+G.

- **MLP-C**: a multilayer perceptron on C, the no-graph control for the GNNs.
- **GNNs**: heterogeneous GraphSAGE [20] (one convolution per relation, mean or sum aggregation), RGCN [16], HAN [17] (claim nodes; meta-paths claim–patient–claim and claim–doctor–claim, with patient, doctor and hospital attributes appended to claim inputs), Simple-HGN [19] and HGT [18]. All share the same per-type input projection, residual connections with layer normalisation, and two-layer classification head.
- **Hybrids**: SAGE+AG and HGT+AG, whose claim nodes also receive the A and G features, testing whether message passing adds information beyond the engineered features. Hybrids reuse the tuned configuration of their base architecture and receive no separate tuning budget.
- **Overlap score**: the degree-normalised triadic closure of the claim, used directly as an unsupervised score.

Tabular models use scikit-learn [34] and graph models PyTorch Geometric [35]. Neural models are trained full-batch with AdamW, class-weighted binary cross-entropy and early stopping on validation average precision (patience 20, at most 200 epochs).

Hyperparameters.

Every method receives the same budget of up to ten configurations (the first being a default), selected by mean validation average precision on two development worlds (generator seeds 901 and 902, $\delta = 0.5$, $s = 0.5$) that are never used for evaluation (Appendix A, Table A2). The development regime ($\delta = 0.5$) is deliberately not one of the evaluation regimes.

4.4 Metrics and statistics

The primary metric is average precision (AP, the area under the precision–recall curve), appropriate under class imbalance [30]. *Ring AP* scores ring claims against legitimate claims (isolated fraud excluded); *isolated AP* is defined analogously. We also report F1 at the threshold that maximises validation F1, ROC-AUC and recall when the top 5% of claims are investigated. Each of the nine regimes is replicated with six independently generated worlds (generator seeds 1–6), 54 worlds in total. Methods are compared within a regime with paired t-tests across worlds; p-values are Holm-adjusted within each pre-specified family of comparisons [32], and Wilcoxon signed-rank tests are reported as a robustness check [31]. Intervals are 95% t-intervals over worlds.

5 Real-World Validation

5.1 Data

We use the public "Healthcare Provider Fraud Detection Analysis" dataset [36] (CC0 public domain), which links Medicare-style inpatient and outpatient claims to beneficiaries and labels each provider as potentially fraudulent or not. Only the labelled training files are used: 558,211 claims (40,474 inpatient, 517,737 outpatient) dated between November 2008 and December 2009, filed by 5,410 providers (506, or 9.4%,

labelled fraudulent) for 138,556 beneficiaries, with 100,737 distinct attending, operating or other physicians. The data documentation does not describe how the labels were obtained, so we treat them as given and discuss this limitation in Section 9.5. Because labels exist only per provider, the prediction unit is the provider; claim-level ring detection cannot be evaluated on these data.

5.2 Features and graph

P (23 provider aggregates): claim volume and inpatient share, beneficiaries and claims per beneficiary, distinct attending and operating physicians, reimbursement totals and log-amount statistics, deductibles, claim duration and length of stay, diagnosis and procedure counts, share of claims with an operating physician, and the age, chronic-condition burden, mortality, renal-disease share, annual reimbursements and geographic spread of the provider's beneficiaries. **G** (8 label-free graph features): for both shared beneficiaries and shared physicians, the number of co-providers, the maximum Jaccard overlap with another provider, the share of the provider's beneficiaries (physicians) seen by other providers, and their mean number of providers. **L** (4 label-neighbourhood features): the share of co-providers that are known fraudulent *training* providers, and an indicator of having any labelled co-provider, through beneficiaries and through physicians. L uses only training labels and never a provider's own label.

The heterogeneous graph has provider, beneficiary and physician nodes, with relations *serves* (provider–beneficiary, 363,300 edges) and *employs* (provider–physician) plus reverse relations. Physicians working for a single provider carry no cross-provider information and are summarised in P rather than kept as nodes, leaving 7,685 shared physicians (16,287 edges). Provider nodes carry P (or P+G for the hybrids), beneficiary nodes 22 demographic and clinical attributes, and physician nodes their claim and provider counts. HAN uses the provider-only meta-paths provider–beneficiary–provider and provider–physician–provider.

5.3 Protocol

All providers' claims are observed and define the graph; labels of validation and test providers are hidden. This matches an audit in which claims are available for every provider but only some providers have been investigated. Providers are assigned to five stratified folds; in rotation k, fold k is the test set, fold $k+1$ the validation set and the remaining three folds the training set, so every provider is tested exactly once. Hyperparameters are selected once, with the same budget of six configurations per method, on a separate stratified 60/20/20 split (seed 1000), and then fixed for the five rotations. Graph models are trained full-batch for at most 100 epochs with early stopping (patience 10) on validation AP. Metrics are AP, ROC-AUC, F1 at the validation-selected threshold, and recall among the top 10% of providers by score (the label prevalence is 9.4%). Methods are compared with paired t-tests over the five rotations (Holm-adjusted); with five rotations these tests have limited power and are interpreted together with effect sizes.

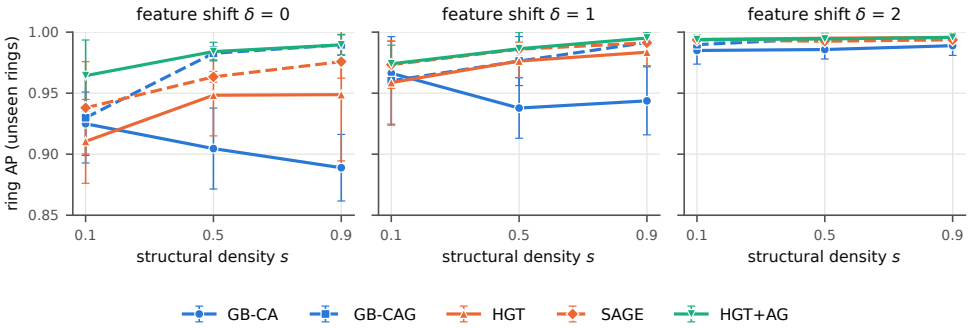


Fig. 3 Ring AP on unseen test rings by feature shift δ and structural density s (mean and 95% t-interval over six worlds). GB-CA: gradient boosting on claim and aggregate features; GB-CAG: plus engineered graph features; HGT+AG: HGT whose claim nodes also receive the A and G features. Feature-only models (GB-C, MLP-C) lie far below the plotted range (Table 1).

6 Results on the Controlled Benchmark

Each test period contains on average 4,924 claims, of which 384 belong to four unseen rings and 133 are isolated fraud, so the chance levels of ring AP and isolated AP are 0.080 and 0.029. Table 1 reports ring AP for every method and regime, Fig. 3 shows the main methods with 95% intervals, and Fig. 4 summarises paired differences. Table 2 pools the paired differences over all 54 worlds.

Table 1 Ring-fraud detection on unseen test rings: average precision of ring versus legitimate claims (chance level 0.08), mean over six worlds per regime. Bold: best per regime. Standard deviations are in the supplementary results files.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.084	0.086	0.087	0.329	0.326	0.334	0.805	0.818	0.809
MLP-C	0.088	0.087	0.095	0.353	0.358	0.360	0.803	0.817	0.808
LR-CA	0.923	0.888	0.868	0.968	0.958	0.952	0.988	0.986	0.987
RF-CA	0.875	0.838	0.844	0.957	0.924	0.941	0.985	0.983	0.986
GB-CA	0.925	0.905	0.889	0.966	0.938	0.944	0.985	0.986	0.989
GB-CAG	0.930	0.983	**0.990**	0.960	0.976	0.992	0.990	0.995	0.994
SAGE	0.938	0.963	0.976	0.973	0.986	0.991	0.994	0.993	0.994
RGCN	0.933	0.953	0.954	0.970	0.982	0.986	**0.995**	0.994	0.996
HAN	0.908	0.906	0.921	0.957	0.964	0.969	0.988	0.989	0.989
Simple-HGN	0.915	0.960	0.957	0.966	0.980	0.982	0.992	0.992	0.993
HGT	0.910	0.948	0.949	0.959	0.976	0.984	0.994	0.995	0.996
SAGE+AG	**0.973**	**0.985**	0.983	0.972	0.984	**0.995**	0.994	**0.996**	**0.997**
HGT+AG	0.964	0.984	0.990	**0.974**	**0.986**	0.995	0.994	0.994	0.996
Overlap score	0.254	0.937	0.985	0.296	0.922	0.989	0.341	0.932	0.981

6.1 Claim features alone do not reveal rings

When ring claims are drawn like legitimate claims ($\delta = 0$), models that see only the claim's own attributes perform at chance (GB-C 0.084–0.087, MLP-C 0.087–0.095). As δ grows their ring AP rises to about 0.33 ($\delta = 1$) and 0.81 ($\delta = 2$). This confirms that the benchmark separates the two sources of evidence as intended.

6.2 Most ring evidence is captured by entity aggregates

Adding per-period aggregates of the claim's patient, doctor and hospital (GB-CA) raises ring AP at $\delta = 0$ to 0.889–0.925. Ring patients file more claims and see more doctors than typical patients, and ring doctors file bursts of claims; these degree and volume signals are present in every regime by design and are captured by simple counts. The remaining gap is where coordination structure matters.

6.3 Coordination structure helps only when the ring block is dense

Engineered graph features add little at low density but substantially at moderate and high density: at $\delta = 0$, GB-CAG improves on GB-CA by +0.005 ($s = 0.1$, n.s.), +0.078 ($s = 0.5$) and +0.101 ($s = 0.9$; Holm $p = 0.003$). The unsupervised overlap score tells the same story: it is weak at $s = 0.1$ (0.254) but reaches 0.937 and 0.985 at $s = 0.5$ and 0.9 without any training. GNNs show the same pattern: SAGE improves on GB-CA by +0.087 at $\delta = 0$, $s = 0.9$ (Holm $p = 0.003$). Pooled over all worlds, every graph-aware method except HAN outperforms GB-CA (GB-CAG +0.031, SAGE +0.031, RGCN +0.026, Simple-HGN +0.024, HGT +0.021, hybrids +0.039; all Holm $p < 0.001$; HAN +0.007, n.s.). When ring claims are strongly anomalous ($\delta = 2$), all methods that see claim and aggregate features exceed 0.98 and the differences become negligible.

6.4 GNNs do not outperform gradient boosting with graph features

The central comparison is between message passing and the same structural information supplied as engineered features. Pooled over the 54 worlds, heterogeneous GraphSAGE matches GB-CAG ($\Delta = -0.0001$, 95% CI $[-0.005, 0.005]$), RGCN is not significantly different (-0.005), and Simple-HGN (-0.008, Holm $p = 0.024$), HGT (-0.011, $[-0.020, -0.002]$, Holm $p = 0.060$) and HAN (-0.024, Holm $p < 0.001$) are lower. Within single regimes, where each test has only six worlds, no GNN is significantly better than GB-CAG in any regime; SAGE, HAN and Simple-HGN are significantly worse at $\delta = 0$, $s = 0.5$, and SAGE and Simple-HGN at $\delta = 0$, $s = 0.9$ (Holm-adjusted within each nine-regime family).

The hybrids, which add message passing on top of the engineered features, gain a small but consistent amount over GB-CAG (SAGE+AG +0.008 $[0.003, 0.013]$, Holm $p = 0.013$; HGT+AG +0.008 $[0.003, 0.013]$, Holm $p = 0.019$), largest in the sparse-structure, no-feature-signal regime ($\delta = 0$, $s = 0.1$: SAGE+AG +0.043, unadjusted $p = 0.011$, Holm $p = 0.61$). Message passing therefore carries a little information beyond our engineered features, but in this benchmark it does not replace them.

Table 2 Paired difference in average precision versus GB-CAG, pooled over all 54 worlds: mean Δ ± half-width of the 95% t-interval, and Holm-adjusted paired t-test p (within each metric).

	Ring AP		Isolated AP		Overall AP	
Method	Δ± CI	p	Δ± CI	p	Δ± CI	p
SAGE	0.000 ± 0.005	0.957	−0.085 ± 0.019	<0.001	−0.007 ± 0.004	0.009
RGCN	−0.005 ± 0.007	0.277	−0.106 ± 0.021	<0.001	−0.013 ± 0.005	<0.001
HAN	−0.024 ± 0.010	<0.001	−0.147 ± 0.024	<0.001	−0.030 ± 0.008	<0.001
Simple-HGN	−0.008 ± 0.006	0.024	−0.094 ± 0.023	<0.001	−0.014 ± 0.005	<0.001
HGT	−0.011 ± 0.009	0.060	−0.118 ± 0.023	<0.001	−0.018 ± 0.007	<0.001
SAGE+AG	+0.008 ± 0.005	0.013	−0.078 ± 0.015	<0.001	0.000 ± 0.003	0.930
HGT+AG	+0.008 ± 0.005	0.019	−0.112 ± 0.018	<0.001	−0.002 ± 0.003	0.388
MLP-C	−0.560 ± 0.079	<0.001	−0.019 ± 0.015	0.015	−0.402 ± 0.062	<0.001

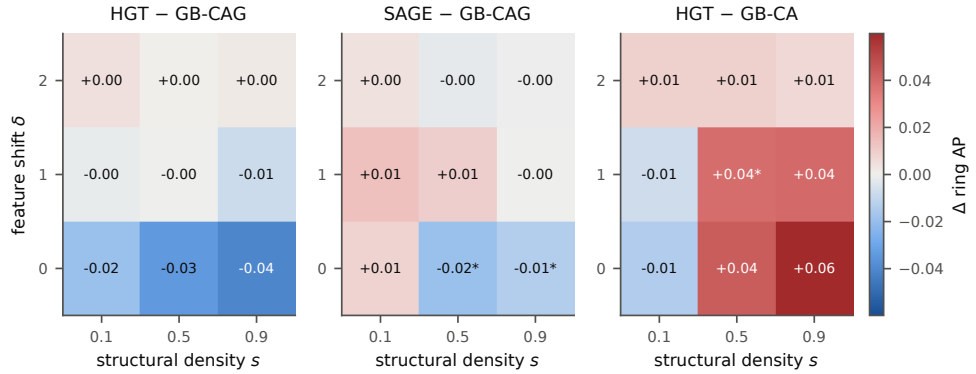


Fig. 4 Mean paired difference in ring AP (six worlds per cell). Left and middle: HGT and SAGE versus gradient boosting with engineered graph features. Right: HGT versus gradient boosting without graph features. *Holm-adjusted $p < 0.05$ within the nine-regime family.

6.5 Message passing dilutes claim-level anomalies

Isolated fraud shows the opposite pattern (Table 3). Every GNN is significantly worse than GB-CAG, by 0.085 (SAGE) to 0.147 (HAN) AP pooled (all Holm $p < 0.001$), with HAN and HGT the weakest. The MLP control, which receives exactly the claim inputs of the GNNs but no messages, comes within 0.019 of GB-CAG. Aggregating a claim's representation with those of its neighbours, which are overwhelmingly legitimate, blurs the claim's own anomaly; the hybrids, whose claim nodes also see the engineered ratios, recover little of the gap. On all fraud combined (overall AP), GB-CAG is better than every pure GNN (by 0.007–0.030, all Holm $p < 0.01$) and indistinguishable from the hybrids.

Table 3 Isolated-fraud detection: average precision of isolated-fraud versus legitimate claims (chance level 0.03), mean over six worlds per regime. Bold: best per regime.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.471	0.473	0.477	0.560	0.556	0.557	0.644	0.649	0.633
MLP-C	0.542	0.539	0.545	**0.624**	**0.614**	0.623	0.651	**0.667**	0.656
LR-CA	0.375	0.313	0.292	0.556	0.549	0.526	**0.672**	0.665	**0.662**
RF-CA	0.551	0.506	0.469	0.586	0.560	0.547	0.626	0.608	0.604
GB-CA	0.502	0.496	0.477	0.586	0.525	0.540	0.643	0.611	0.627
GB-CAG	**0.575**	**0.613**	**0.628**	0.604	0.605	**0.629**	0.660	0.662	0.655
SAGE	0.447	0.448	0.509	0.536	0.548	0.555	0.615	0.605	0.602
RGCN	0.428	0.425	0.474	0.499	0.516	0.512	0.609	0.602	0.607
HAN	0.375	0.371	0.408	0.472	0.467	0.466	0.590	0.576	0.581
Simple-HGN	0.408	0.401	0.484	0.537	0.518	0.534	0.638	0.639	0.625
HGT	0.344	0.379	0.455	0.500	0.498	0.537	0.622	0.626	0.605
SAGE+AG	0.457	0.503	0.586	0.493	0.548	0.582	0.582	0.584	0.598
HGT+AG	0.378	0.454	0.553	0.444	0.498	0.566	0.549	0.576	0.604
Overlap score	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027

6.6 Cost

Median training time per world (Intel Core i7-1068NG7 laptop CPU, two threads per job with several jobs in parallel, so times are indicative) was 3 s for GB-CAG, 39 s for SAGE, 56 s for RGCN, 86 s for HAN, 126 s for HGT and 144 s for Simple-HGN; scoring a test period took at most 0.25 s for any method.

7 Mechanism and Memorisation

7.1 What the models rely on

Table 4 and Fig. 5 report ablations in the regime where structure matters most ($\delta = 0$, $s = 0.9$). Degree-preserving rewiring—permuting which doctor bills each claim within a period, which keeps every patient's and doctor's claim volume but destroys the ring's co-occurrence block—reduces every graph-aware method to the same level (0.757–0.770), close to gradient boosting without graph features after rewiring (0.748). The GNNs therefore do exploit coordination structure, and once it is removed nothing distinguishes them from a model that sees volume alone.

Depth matters: with a single message-passing layer, ring AP falls by 0.135 (SAGE) and 0.161 (HGT); two and three layers perform alike. Removing the direct patient–doctor relation costs little (-0.016 and -0.047, not significant after correction), because two-hop paths through claim nodes carry the same information. The largest effect is specific to HGT: without degree encodings its ring AP drops from 0.949 to 0.569 (-0.379, Holm $p = 0.001$), whereas sum-aggregating SAGE changes little (-0.029, n.s.). Softmax attention normalises over neighbours and so cannot count them; supplying counts as input features, or aggregating by summation, restores the signal [21].

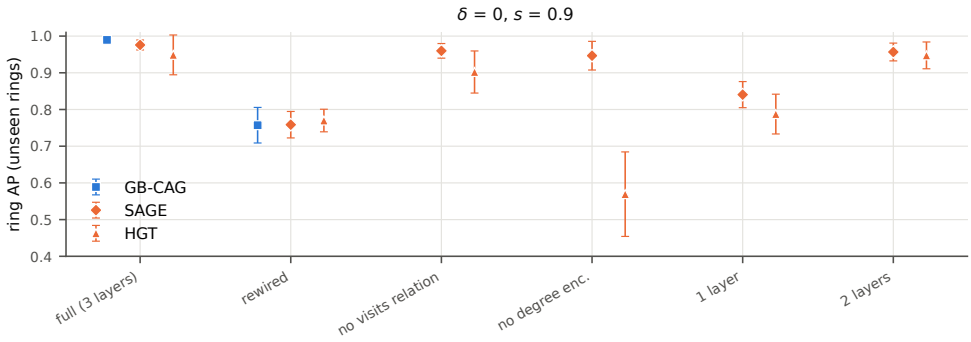


Fig. 5 Mechanism ablations at $\delta = 0$, $s = 0.9$ (mean and 95% t-interval over six worlds). Rewiring preserves every node's degree but destroys the ring block.

Table 4 Mechanism ablations at $\delta = 0$, $s = 0.9$ (six worlds): ring AP and paired change from the full model (Holm-adjusted p over all ablations). The tuned GNNs use three layers.

Method	Variant	Ring AP	Δ vs. full
GB-CA	Full model	0.889 ± 0.026	–
	Degree-preserving rewiring	0.748 ± 0.027	−0.141; $p = 0.009$
	+ ring identifier (legacy leak)	1.000 ± 0.000	+0.111; $p = 0.001$
GB-CAG	Full model	0.990 ± 0.008	–
	Degree-preserving rewiring	0.757 ± 0.046	−0.232; $p < 0.001$
SAGE	Full model	0.976 ± 0.013	–
	Degree-preserving rewiring	0.759 ± 0.034	−0.217; $p < 0.001$
	No patient–doctor relation	0.960 ± 0.019	−0.016; $p = 0.138$
	No degree encodings	0.947 ± 0.037	−0.029; $p = 0.232$
	1 message-passing layer	0.841 ± 0.034	−0.135; $p < 0.001$
	2 layers	0.957 ± 0.023	−0.019; $p = 0.138$
HGT	Full model	0.949 ± 0.052	–
	Degree-preserving rewiring	0.770 ± 0.029	−0.179; $p = 0.014$
	No patient–doctor relation	0.902 ± 0.055	−0.047; $p = 0.138$
	No degree encodings	0.569 ± 0.110	−0.379; $p = 0.001$
	1 message-passing layer	0.788 ± 0.052	−0.161; $p = 0.009$
	2 layers	0.947 ± 0.035	−0.001; $p = 0.880$
	No reverse relations (legacy graph)	0.772 ± 0.024	−0.177; $p = 0.010$

7.2 Two evaluation pitfalls, quantified

The same experiment quantifies two defects that are easy to introduce. Adding the generator's ring identifier to the feature set—a column that is non-zero only for ring claims—raises gradient boosting from 0.889 to a perfect 1.000. Building the heterogeneous graph with one-directional relations only (patient→claim→doctor→hospital) leaves claim nodes reachable only from patients, so doctor context never reaches them; HGT then loses 0.177 AP. Both defects produce plausible-looking numbers and neither is caught by aggregate metrics alone; the tests listed in Section 10 detect both.

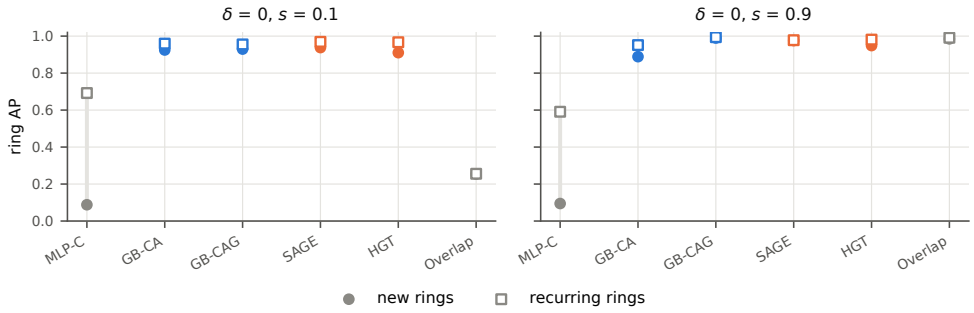


Fig. 6 Ring AP on new versus recurring rings ($\delta = 0$, six worlds).

7.3 Recognising known members is not detecting rings

Table 5 and Fig. 6 compare new rings (emergent protocol) with rings whose members recur from training (persistent protocol). The MLP control, which sees only the claim and the static attributes of its patient, doctor and hospital, rises from chance to 0.692 ($s = 0.1$) and 0.591 ($s = 0.9$) when rings recur: it identifies the members from their attributes. HGT gains 0.056 at $s = 0.1$ (95% CI $[0.029, 0.084]$), and gradient boosting with aggregates gains 0.035–0.063, while the unsupervised overlap score, which cannot memorise, is unchanged. Evaluations in which test rings share members with training rings therefore overstate how well a model detects coordination, by an amount that differs between methods and can reorder them.

Table 5 Ring AP when test rings are new (emergent) versus recurring members of training rings (persistent), $\delta = 0$, six worlds each; gain = persistent − emergent (paired over worlds, 95% t-interval). The overlap score is unsupervised and cannot memorise.

	$s = 0.1$			$s = 0.9$		
Method	new	recurring	gain	new	recurring	gain
MLP-C	0.088	0.692	+0.604 [+0.523, +0.685]	0.095	0.591	+0.496 [+0.416, +0.577]
GB-CA	0.925	0.959	+0.035 [+0.014, +0.055]	0.889	0.952	+0.063 [+0.034, +0.092]
GB-CAG	0.930	0.955	+0.025 [−0.012, +0.062]	0.990	0.994	+0.005 [−0.002, +0.012]
SAGE	0.938	0.969	+0.031 [−0.019, +0.081]	0.976	0.978	+0.002 [−0.015, +0.020]
HGT	0.910	0.967	+0.056 [+0.029, +0.084]	0.949	0.981	+0.032 [−0.017, +0.082]
Overlap score	0.254	0.256	+0.002 [−0.137, +0.142]	0.985	0.991	+0.006 [−0.009, +0.020]

8 Results on Real Data

Table 6 and Fig. 7 report provider-level results over the five rotations (1,082 test providers each, about 101 fraudulent). Gradient boosting on provider aggregates (GB-P) has the highest AP (0.704 ± 0.045). Adding label-free graph features (GB-PG, -0.009, 95% CI $[-0.029, 0.011]$) or graph and label-neighbourhood features (GB-PGL, -0.007 $[-0.021, 0.007]$) does not improve it. Every GNN has lower AP than GB-P, by -0.029 (HGT) to -0.083 (RGCN); with five rotations most of these differences are not significant after Holm adjustment over all 24 comparisons, the exception being SAGE+G (-0.051, Holm $p = 0.021$). The hybrids, which give the GNN the provider and graph features, are no better than the plain GNNs. ROC-AUC lies between 0.927 and 0.942 for all methods and recall among the top 10% of providers between 0.59 and 0.64, so the methods rank providers similarly; the AP differences come from the top of the ranking. Under the same parallel load as before, median training time per rotation was 2–15 minutes for the GNNs (range 1.4–24 minutes) against 7 s for GB-P.

Table 6 Provider-level fraud detection on the public Medicare-style provider dataset (5-fold rotation; mean ± s.d. over folds). F1 at the validation-selected threshold; Recall@10% = share of fraudulent providers among the top 10% scored.

Method	AP	ROC-AUC	F1	Recall@10%
LR-P	0.678 ± 0.041	0.940 ± 0.017	0.617 ± 0.030	0.644 ± 0.023
RF-P	0.681 ± 0.049	0.942 ± 0.017	0.624 ± 0.031	0.644 ± 0.025
GB-P	0.704 ± 0.045	0.939 ± 0.020	0.624 ± 0.028	0.636 ± 0.027
GB-PG	0.695 ± 0.040	0.937 ± 0.019	0.600 ± 0.043	0.642 ± 0.024
GB-PGL	0.697 ± 0.042	0.937 ± 0.019	0.595 ± 0.045	0.634 ± 0.019
MLP-P	0.685 ± 0.052	0.940 ± 0.019	0.613 ± 0.028	0.642 ± 0.032
SAGE	0.664 ± 0.047	0.938 ± 0.018	0.605 ± 0.036	0.624 ± 0.046
RGCN	0.621 ± 0.054	0.927 ± 0.019	0.578 ± 0.029	0.593 ± 0.029
HAN	0.666 ± 0.046	0.938 ± 0.018	0.583 ± 0.028	0.636 ± 0.045
Simple-HGN	0.670 ± 0.057	0.939 ± 0.020	0.600 ± 0.037	0.630 ± 0.036
HGT	0.676 ± 0.058	0.939 ± 0.019	0.613 ± 0.051	0.640 ± 0.051
SAGE+G	0.653 ± 0.048	0.937 ± 0.018	0.612 ± 0.040	0.623 ± 0.031
HGT+G	0.648 ± 0.032	0.933 ± 0.014	0.589 ± 0.022	0.615 ± 0.035

The graph statistics explain why. Almost every provider (99.5%) shares beneficiaries with other providers (median 30 co-providers), but labels are not assortative through this relation: on average 23.9% of a fraudulent provider's co-providers are labelled fraudulent, against 25.3% for a legitimate provider (both well above the 9.4% base rate because shared beneficiaries concentrate on large providers). Shared physicians show mild assortativity (19.1% versus 14.2%) but connect only 73% of providers, with a median of two co-providers. Fraudulent providers are, however, much larger (median 156 versus 27 claims), a volume signal that provider aggregates capture directly. In the terms of our benchmark, these data resemble the low-density regime, where structure added nothing beyond per-entity aggregates. The labels mark potentially fraudulent providers without identifying rings, so the absence of a graph benefit here does not show that rings are absent from Medicare data; it shows that this public labelling contains no exploitable relational signal for the models tested.

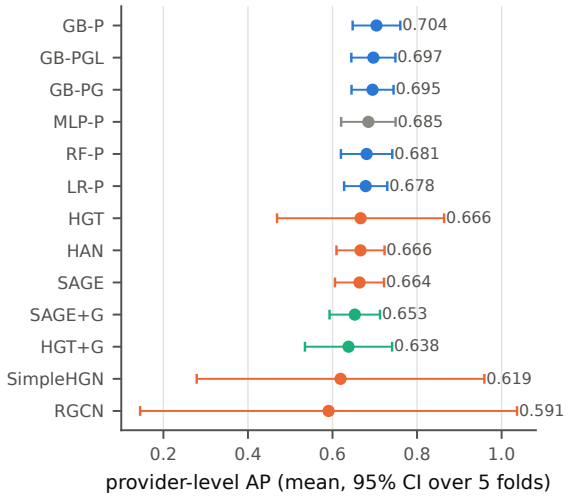


Fig. 7 Provider-level AP on the real data (mean and 95% t-interval over five rotations). Blue: tabular; orange: GNN; aqua: hybrid; gray: MLP.

9 Discussion

9.1 When does relational structure help?

Three conditions emerge from the controlled study. First, coordination must leave a structural trace beyond volume: at low block density, rings were detected mainly through the elevated utilisation of their members, which per-entity counts capture. Second, the trace must be dense enough to distinguish a ring from ordinary shared care: graph information added 0.04–0.10 AP at moderate and high density (most when ring claims looked legitimate) but nothing measurable at low density. Third, the claims must not already be anomalous on their own: when ring claims were strongly upcoded ($\delta = 2$), every model with claim and aggregate features exceeded 0.98 AP and structure became redundant.

9.2 Why GNNs did not beat engineered graph features

When structure did help, gradient boosting with a handful of shared-patient statistics captured it as well as or better than the best GNN. The mechanism analyses suggest why. The useful signal here is largely *counting*—how many of a patient's doctors also share other patients—and attention- or mean-based aggregators cannot count: HGT lost most of its ring detection without degree encodings, whereas sum-aggregating GraphSAGE barely changed (Section 7). This mirrors the expressiveness argument for sum aggregation [21]. GNNs also dilute claim-level anomalies through neighbourhood averaging, which made them markedly worse on isolated fraud, and they readily memorise the attributes of training-ring members, which inflates performance when rings recur but does not transfer to new rings. Hybrids that combine message passing with engineered features recovered a small, consistent gain over gradient boosting on

ring fraud, indicating that the GNN extracted some structure our features miss, at 9–40 times the training cost.

These findings are consistent with evidence from other domains that tree ensembles with neighbourhood features are strong graph anomaly detectors [22, 23], and extend it to heterogeneous patient–provider claim graphs with controlled typologies.

9.3 Real data

On the public provider-labelled data, gradient boosting on provider aggregates was the strongest method, and neither engineered graph features, label-neighbourhood features nor any GNN improved on it. The labels are not assortative through shared beneficiaries, and fraudulent providers differ mainly in volume, so the data sit in the regime where our benchmark also found no benefit from structure. This is consistent with the controlled results but is not an independent confirmation of the dense-ring regime: validating that regime requires real data with labelled collusion, which are not publicly available.

9.4 Practical recommendations

- Start with gradient boosting on claim features, per-entity aggregates and a small set of shared-patient graph features; on our benchmark this was never significantly worse than any pure GNN, within 0.01 AP of the hybrids on average, and far cheaper.
- Evaluate on rings (or providers) that do not appear in training, and compare against a no-graph control with identical inputs; otherwise memorisation of known members can masquerade as relational learning.
- If a GNN is used, include reverse relations, verify that information reaches the classified nodes, and give attention-based architectures structural (degree) encodings or prefer sum aggregation.
- Keep isolated-fraud detection on a claim-level model; message passing degraded it consistently.

9.5 Limitations

The controlled findings hold for our generator. Its distributions are simple, labels are noise-free and immediate, rings are static during their active period and of a single size, and ring prevalence (about 8% of test-period claims) is far above realistic levels so that each world has enough positives for stable estimates. Ring members also carry a volume signal in every regime, so our regimes vary the *additional* structural evidence rather than all structural evidence. We tuned each method once, on development worlds from a single regime ($\delta = 0.5$, $s = 0.5$); per-regime tuning could change small differences. Six worlds per regime give limited power within a regime, which is why we emphasise pooled comparisons. Graph models were trained full-batch on graphs of about 11,000 nodes per period; larger graphs would need sampling-based training [20]. On the real data, labels are per provider and of undocumented provenance, the observation window is a single year, and hyperparameters were selected on a split whose validation providers overlap the evaluation test folds; this overlap

affects all methods equally but may favour each slightly. Finally, the benchmark does not model adaptive fraudsters, temporal ring dynamics, which temporal graph models target [38], or realistic coding semantics such as those of synthetic health records [37].

10 Reproducibility

All results are produced by one released package from raw inputs. Every run writes a record containing the git commit, a hash of its configuration, the hashes of the generated claim and ground-truth tables, the library versions and the per-claim (or per-provider) test predictions; the tables and figures of this paper are generated from these records by a single script. Automated tests, run in continuous integration, check that (i) no ground-truth column reaches any feature matrix and features are invariant to permuting the labels; (ii) a period's features and graph are unchanged when claims of other periods are perturbed; (iii) emergent rings in different splits share no doctor or patient; (iv) at $\delta = 0$ ring claims are distributed like legitimate claims; (v) features, graphs and labels stay aligned by claim identifier; and (vi) for every GNN, perturbing doctor features changes claim logits, so relational information actually reaches the classified nodes. A checklist mapping this study to the REFORMS recommendations [40] and a datasheet for the generator [39] are included in the repository.

11 Conclusion

We asked when relational structure helps healthcare fraud detection and answered it with a benchmark that varies the feature-level and structural signal of provider rings independently, tests on rings never seen in training, and checks the conclusions on public real data. Structure helped only when rings formed dense doctor–patient blocks, and in that regime a gradient-boosting model with a handful of engineered shared-patient features matched or outperformed five heterogeneous GNNs at a fraction of the cost; adding message passing on top of those features gained little. GNNs generalised to new rings only with reverse relations, sufficient depth and counting ability, degraded isolated-fraud detection, and could memorise known ring members in ways that inflate conventional evaluations. On provider-labelled real data no graph-based method improved on tabular aggregates. We recommend that graph models for claims fraud be justified against a graph-augmented tabular baseline and a no-graph control, on entities unseen in training. The benchmark, code and tests are released to support such comparisons; the most valuable next step is real data with labelled collusive schemes.

Supplementary information. The repository contains the per-world results (including standard deviations, F1, ROC-AUC and recall at a fixed investigation budget for every regime and method), the tuning logs and the test predictions underlying every table and figure.

Declarations

- **Funding.** The authors did not receive support from any organization for the submitted work.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable: the study uses synthetic data and a public, de-identified dataset released under a CC0 licence; no human participants were involved.
- **Consent for publication.** Not applicable.
- **Data availability.** The synthetic benchmark is generated by the released code from fixed seeds. The real-world data are available from Kaggle [36] under CC0; file checksums are listed in the repository.
- **Code availability.** Code, configurations, tests and result records are available at https://github.com/akshay-hudev/Fraud-Guard (directory `research/`) under the MIT licence. An archived release with a DOI will be provided upon acceptance.
- **Author contribution.** H M Akshay: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization and writing—original draft. Gaurav S B, Eshita Narayana and Harshita Srivastava: methodology, investigation, formal analysis and writing—original draft. Malini M Patil: supervision and writing—review and editing. All authors read and approved the final manuscript.

Appendix A Generator parameters and hyperparameters

Table A1 Generator parameters (defaults in `research/synth/generator.py`).

Parameter	Value
Patients / doctors / hospitals / regions / specialties	6,000 / 300 / 40 / 5 / 8
Audit periods	4 × 6 months (train: 1–2, validation: 3, test: 4)
Legitimate claims per patient	Poisson(gamma rate; mean 3, shape 1.5) over 24 months
Share of legitimate claims with a regular doctor	0.85 (1–3 regular doctors per patient, same region)
Claim amount	log-normal: specialty mean + 0.15 (procedures − 1) + 0.8 inpatient, s.d. 0.6
Isolated fraud	3% of legitimate volume; feature shift 2.0
Ring size	4 doctors, 12 recruited patients (utilisation-weighted)
Ring volume	16 claims per month during activity; monthly burst days
Rings per period	4 (emergent: new members each period; persistent: same 4 rings)
Feature shift δ	0, 1, 2 (development worlds: 0.5)
Structural density s	0.1, 0.5, 0.9 (development worlds: 0.5)
Worlds per regime	6 (seeds 1–6); development worlds: seeds 901, 902

Table A2 Selected hyperparameters (equal budget per method; hybrids inherit their base configuration; methods sharing a search space were offered the same candidate configurations). Synthetic: tuned on development worlds; real data: tuned on a separate split.

Method	Synthetic benchmark	Real data
LR	C=100	C=1
RF	max depth=16; max features=sqrt; min samples leaf=3; n estimators=300	max depth=16; max features=0.3; min samples leaf=10; n estimators=300
GB (C)	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=10	–
GB (C+A / P)	l2=1; learning rate=0.1; max iter=200; max leaf nodes=63; min samples leaf=50	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (C+A+G / P+G)	l2=1; learning rate=0.1; max iter=400; max leaf nodes=15; min samples leaf=20	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (P+G+L)	–	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
MLP	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05
SAGE	aggr=sum; dropout=0.1; hidden=32; layers=3; lr=0.003; weight decay=0.001	aggr=mean; dropout=0.3; hidden=64; layers=3; lr=0.003; weight decay=0.0001
RGCN	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.003; weight decay=0.001	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.01; weight decay=0.001
HAN	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=2; hidden=64; layers=2; lr=0.003; weight decay=0.0001
Simple-HGN	dropout=0.3; heads=4; hidden=64; layers=3; lr=0.003; weight decay=0.0001	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001
HGT	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001

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